

KHRABROVO, Russian Federation

WMO: 267020

Lat: **54.890N**Lon: **20.593E**Elev: **13**StdP: **101.17**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.4	-13.1	-18.2	0.8	-14.9	-15.4	1.0	-11.9	13.8	3.0	11.7	3.4	1.9	110	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	28.3	20.2	26.3	19.1	24.7	18.3	21.4	26.2	20.3	24.7	19.2	23.3	3.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	14.6	23.9	18.8	13.6	22.5	17.8	12.8	21.4	62.4	26.0	58.2	24.7	54.6	23.2	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.7	8.3	7.4	-19.7	31.8	4.4	1.9	-22.9	33.2	-25.5	34.3	-28.0	35.4	-31.2	36.8
			-19.8	23.8	4.2	1.4	-22.8	24.7	-25.3	25.5	-27.7	26.3	-30.7	27.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-1.8	-0.9	2.2	7.6	12.4	15.7	18.4	17.9	13.6	8.5	4.0	0.2
	DBStd	8.15	5.56	5.34	3.96	4.03	3.86	2.99	2.86	2.63	2.92	3.55	3.93	5.09
	HDD10.0	1590	367	306	243	94	17	0	0	0	6	71	182	304
	HDD18.3	3784	626	539	501	322	188	90	35	40	144	304	431	562
	CDD10.0	932	0	0	1	22	92	172	261	246	113	24	1	0
	CDD18.3	82	0	0	0	0	5	11	37	27	2	0	0	0
	CDH23.3	683	0	0	0	11	60	114	281	201	17	0	0	0
	CDH26.7	153	0	0	0	1	7	26	79	39	2	0	0	0
Wind	WSAvg	3.2	3.7	3.6	3.5	3.2	3.0	3.0	2.8	2.6	2.8	3.2	3.5	3.7
Precipitation	PrecAvg	782	61	49	47	38	52	69	89	90	72	77	75	65
	PrecMax	1094	146	80	102	78	110	129	199	192	147	172	156	98
	PrecMin	573	10	13	6	1	16	10	10	5	16	14	25	37
	PrecStd	108	29	18	26	19	22	27	47	45	35	41	35	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	9.5	15.3	24.9	27.8	29.9	31.8	30.2	25.5	19.2	12.8	10.1
		MCWB	7.4	6.5	8.8	16.0	18.5	21.3	21.6	21.4	18.2	14.8	10.8	8.7
	2%	DB	7.0	7.9	11.9	20.8	25.0	26.7	28.9	27.2	22.4	17.0	11.1	8.4
		MCWB	5.9	6.0	7.6	13.3	17.1	19.3	20.7	19.6	17.1	13.6	9.5	7.4
	5%	DB	5.8	6.2	9.2	17.3	22.2	24.0	26.3	25.4	20.5	15.1	10.0	7.2
		MCWB	4.7	4.9	6.4	11.0	15.6	17.6	19.7	18.8	15.9	12.8	8.8	6.3
	10%	DB	4.4	5.1	7.6	14.8	19.8	21.9	24.1	23.8	18.9	13.8	9.0	6.1
		MCWB	3.6	4.0	5.3	9.8	14.2	16.5	18.8	18.2	15.1	12.0	7.9	5.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	7.5	10.2	16.3	19.9	22.1	23.6	22.7	19.5	15.7	11.3	9.2
		MCDB	8.4	8.4	13.3	24.1	26.1	27.7	29.0	27.5	23.7	18.1	12.4	9.8
	2%	WB	5.9	6.1	8.2	13.6	17.9	20.0	21.8	21.0	17.8	14.2	9.9	7.6
		MCDB	6.7	7.3	10.7	19.6	23.5	25.4	26.6	25.6	21.0	16.2	10.8	8.2
	5%	WB	4.6	5.1	6.7	11.8	16.4	18.5	20.5	19.8	16.6	13.1	8.9	6.4
		MCDB	5.4	6.1	8.8	16.3	21.2	22.9	24.8	23.7	19.6	14.8	9.8	7.1
	10%	WB	3.6	4.0	5.4	10.2	14.8	17.2	19.5	18.8	15.6	12.2	7.9	5.2
		MCDB	4.3	4.9	7.1	13.8	18.9	20.8	23.5	22.3	18.2	13.7	8.8	6.0
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.4	7.1	10.0	10.5	9.7	9.3	9.6	8.8	7.0	4.6	4.4
		MCDBR	4.5	5.4	10.3	15.1	14.4	14.0	13.0	12.8	11.9	9.2	5.7	4.4
		MCWBR	3.9	4.2	6.9	8.5	7.6	7.4	6.3	6.4	7.0	6.2	4.7	4.0
	5% WB	MCDBR	4.3	5.0	9.5	14.0	13.3	12.5	11.8	11.1	10.6	8.1	5.2	4.4
		MCWBR	3.9	4.3	6.8	8.3	7.6	7.3	6.4	6.2	6.6	6.0	4.6	4.2
Clear-Sky Solar Irradiance	taub	0.311	0.338	0.367	0.408	0.395	0.395	0.426	0.419	0.378	0.359	0.346	0.309	
	taud	2.254	2.265	2.262	2.218	2.326	2.362	2.292	2.311	2.400	2.410	2.299	2.253	
	Ebn at Noon	611	730	796	813	849	853	814	795	782	705	561	523	
	Edh at Noon	59	82	103	124	117	115	121	112	90	72	57	48	
All-Sky Solar Radiation	RadAvg	0.50	1.21	2.49	4.16	5.46	5.66	5.20	4.35	3.08	1.54	0.57	0.35	
	RadStd	0.08	0.20	0.31	0.52	0.49	0.42	0.56	0.44	0.35	0.24	0.07	0.04	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-140	+63	N/A

Nomenclature: See separate page